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Education

Harvard University

Ph.D. Economics, 2022 to 2027 (expected)

Yale University

B.A. Statistics and Data Science, 2020

B.A. Ethics, Politics, and Economics, 2020

Summa cum laude, distinction in both majors

Fields

Primary: Econometrics

Secondary: Machine Learning Methods for Economics

References

Isaiah Andrews
iandrews@mit.edu

Melissa Dell
melissadell@fas.harvard.edu

Neil Shephard
shephard@fas.harvard.edu

Davide Viviano
dviviano@fas.harvard.edu

Fellowships & Awards

PhD Fellowship Finalist, Two Sigma, 2026

Presidential Scholarship, Harvard University, 2022

Undergraduate Thesis Prize Winner, Forethought Foundation for Global Priorities Research, 2020

Phi Beta Kappa, Yale University, 2020

Teaching

Causal Inference in Economics (PhD, Spring 2025, 2026, Davide Viviano)

Time Series (PhD, Fall 2024, 2025, Neil Shephard)

Unleashing Novel Data at Scale (PhD, Spring 2021, Melissa Dell)

Employment

Harvard University, Data Science Fellow in Deep Learning, 2020 to 2022

Job Market Paper

Making Interpretable Discoveries from Unstructured Data: A High-Dimensional Multiple Hypothesis Testing Approach

Social scientists are increasingly turning to unstructured datasets to unlock new empirical insights, e.g., estimating descriptive statistics of or causal effects on quantitative measures derived from text, audio, or video data. In many settings, unsupervised analysis is of primary interest, in that the researcher does not want to (or cannot) manually pre-specify all important aspects of the unstructured data to measure; they are interested in “discovery.” This paper proposes a general and flexible framework for pursuing such discovery from unstructured data in a statistically principled way. The framework leverages recent methods from the literature on AI interpretability to map unstructured data points to high-dimensional, sparse, and interpretable “concept embeddings”; computes statistics from these concept embeddings for testing interpretable, concept-by-concept hypotheses; performs selective inference on these hypotheses using algorithms validated by new results in high-dimensional central limit theory, producing a selected set (“discoveries”); and both generates and evaluates human-interpretable natural language descriptions of these discoveries. The proposed framework has few researcher degrees of freedom, is robust to data snooping and other post-selection inference concerns, and facilitates fast and inexpensive sensitivity analysis and

replication. Applications to recent descriptive and causal analyses of unstructured data in empirical economics are explored.

Publications

Carlson, Jacob, Tom Bryan, and Melissa Dell. “Efficient OCR for Building a Diverse Digital History.” *ACL* (2024). **Oral presentation.**

Dell, Melissa, Jacob Carlson, Tom Bryan, Emily Silcock, Abhishek Arora, Zejiang Shen, Luca D'Amico-Wong, Quan Le, Pablo Querubin, Leander Heldring. “American Stories: A Large-Scale Structured Text Dataset of Historical U.S. Newspapers.” *NeurIPS* Datasets and Benchmarks Track (2023).

Bryan, Tom, Jacob Carlson, Abhishek Arora, Melissa Dell. “EfficientOCR: An Extensible, Open-Source Package for Efficiently Digitizing World Knowledge.” *EMNLP* System Demonstrations (2023).

Carlson, Jacob, Trevor Incerti, and P.M. Aronow. “Dyadic Clustering in International Relations.” *Political Analysis* (2024).

Shen, Zejiang, Ruochen Zhang, Melissa Dell, Benjamin Lee, Jacob Carlson, and Weining Li. “LayoutParser: A Unified Toolkit for Deep Learning Based Document Image Analysis.” *ICDAR* (2021). **Oral presentation.**

Working Papers

Carlson, Jacob and Neil Shephard. “When are time series predictions causal? The potential system and dynamic causal effects.” *Working paper* (2026).

Carlson, Jacob and Melissa Dell. “A Unifying Framework for Robust and Efficient Inference with Unstructured Data.” *Working paper* (2025).

Andrews, Isaiah, Harvey Barnhard, and Jacob Carlson. “True and Pseudo-True Parameters.” *Working paper* (2024). Conditionally accepted, *Econometric Theory*.

Seminars & Conferences

2026: ESIF Economics and AI+ML Meeting (Cornell, June); AI and Economics Summer Conference (Chicago Booth, August)

2022: ASSA Annual Meeting (Virtual, January)

Referee Service

Journal of Political Economy

Languages

English (Native)

Personal Information

Citizenship: U.S.A.